Build a small "**Task management system**" with role-based access and audit logging. The goal is to evaluate your **architecture decisions, code structure, and problem-solving approach**, not just feature completion.

Features

User Management

- System must have 2 role support
    1. **Admin**
        - Create, update, delete tasks
        - Assign tasks to users
        - View audit logs
    2. **User**
        - View assigned tasks
        - Update task status

Authentication

- Simple authentication (JWT/session)
- Predefined users, no registration flow is needed:
    - Admin user
    - Normal user

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Task Management

Each task should include:

- Title
- Description
- Status (PENDING, PROCESSING, DONE)
- Assigned user
- Created & updated timestamps

Audit Log (Important)

All important actions must be logged:

- Task creation, update, deletion
- Status changes
- Assignment changes

Each log should include:

- Actor (who performed the action)
- Action type
- Target entity (task)
- Relevant data (before/after or summary)

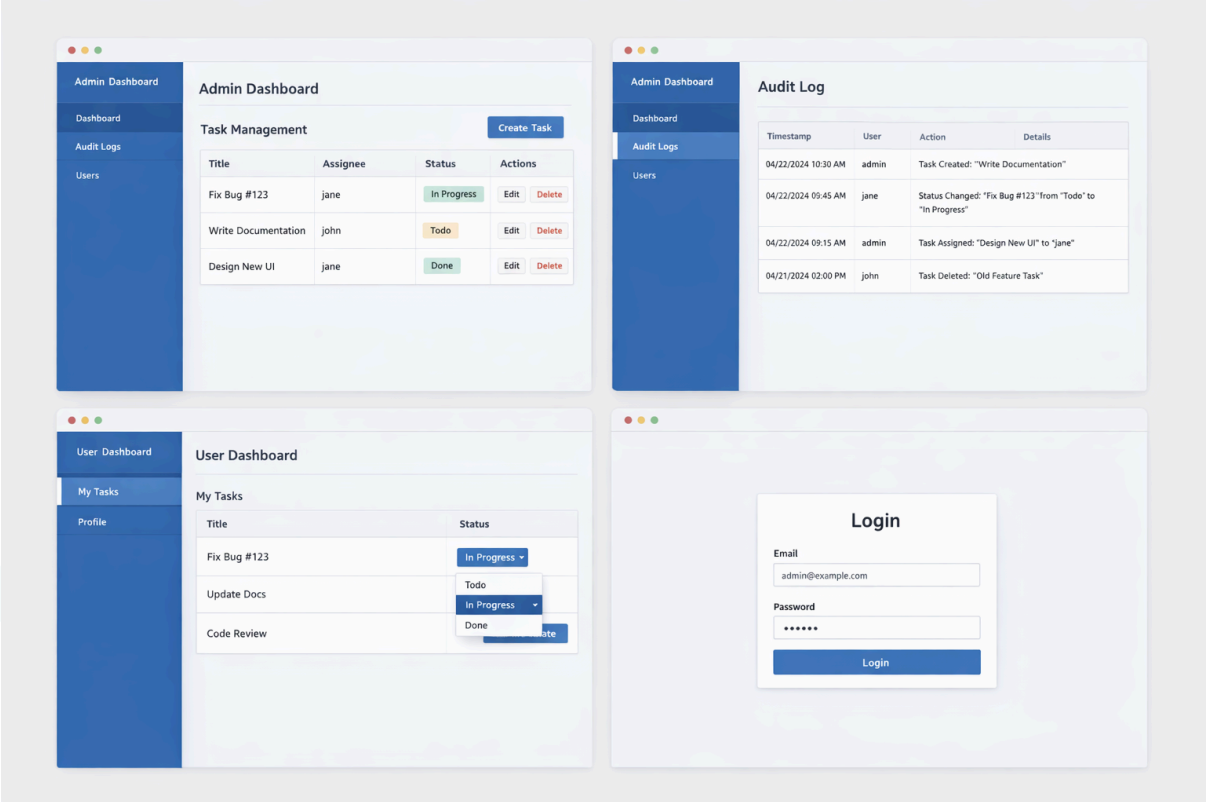

Fig: It's a sample UI to give an idea of the features

Technology stack:

- For frontend use **Nextjs** or any reactjs based framework
- Backend should be **Nestjs** based
- Use **postgreSQL** database with any ORM

Evaluation Criteria

- Code structure & maintainability
- Architecture decisions
- Data modeling & API design
- Audit log implementation quality
- Completeness & usability
- Clarity of explanation (video)

Notes

- Keep it simple but well-structured
- Focus on clean design over adding extra features
- You are encouraged to use AI tools, but be prepared to explain your decisions

Deliverables

1. Source Code (GitHub repo) links
2. Docker setup (**"docker compose up"** should run everything) inside the repo
3. Demo credentials (admin + user) inside the repo. (A readme file)
4. Short video (5 min max) demonstrating the frontend UI and its features, code structure, database schema/table
5. Major AI prompts if AI is used. Explain what prompt you gave and why you used AI for this task.

Last submission time: 9th April, 2026 at 10.00 AM

****No applications will be accepted after this time*

Submission link: https://forms.gle/b2JVXeBHeWiSavKC8